

Action Scripts Overview

According to the data gathered from various DELMIA Apriso projects, most DFCs that contain business logic use one of these two functions: [User Formula](#) or [SQL Query](#). Action Scripts can be used as replacements for these types of DFCs, as they enable getting rid of design and runtime overhead during execution.

Action Scripts are simple pieces of code that can replace DFCs in certain situations. The language used for Action Scripts is TypeScript. This is a Microsoft programming language and a strict superset of JavaScript with optional static typing and class-based object-oriented programming.

Action Scripts are available only at the level of Screens and Views, but they do not support display output. They can be used to validate the data in forms, parsing, or other strictly back-end actions. Even if there is a user interface displayed in an Action Script, it will not be supported. Each Action Script "file" is kept in the DELMIA Apriso database, so it is not a physical file.

Action Scripts Usage

It is possible to use Action Scripts in the following places:

- ▶ Screen "On Load" Property
- ▶ Screen "On Initialize" Property
- ▶ View Actions "On Action" Property
- ▶ View Forms "On Change" Property
- ▶ Offline Module "Initialization Script" Property

It is possible to use (link) only Action Script Functions that belong to Action Scripts in the Active, Prototype or Development in Progress status and in the default revision. To use Action Script Functions in runtime, they need to be compiled.

Whenever an Action Script is used, it becomes a part of the DFC and is included in the transaction stack.

Limitations

- ▶ Action Scripts are not [moved](#) with entities that use them. You need to link the Action Script manually after moving the entity.
- ▶ After duplicating a project with a linked [initialization script](#), the link is removed and the status of the script changes to Design in Progress. You need to manually change the script status to Prototype and link it again.

Action Scripts on the Client Side and Server Side

Action Scripts can be run on the [client side](#) (without a call to the server) or on the server side (each Action Script execution reaches the server for additional data). Action Scripts that are executed only in a Web browser are usually simple validation scripts of data on the displayed Web page, or they are data modifications or conversions.

It is impossible to manually specify that Action Scripts should be executed without callback to the server. Depending on whether the Action Script is client-side or server-side, the system decides by itself what to do during the compilation of the Action Script.

To make sure that an Action Script is client-side, use only the classes that perform the following:

- ▶ Call only methods and functions from standard JavaScript objects (for example, Number, String, Array, Date, Math, RegExp)
- ▶ Use only global functions and properties (for example, isNaN, decodeURI, encodeURI, isFinite, parseFloat, parseInt)
- ▶ Use only functions for raising errors or logging debug, information, and error messages from the [DELMIA Apriso Action Script API](#)